F-35 RC Airframe — Engineering Analysis

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Summary

The airframe prints in about 210 g of PLA for $2–6, but at an estimated 420 g flying weight it stalls at roughly 30–34 mph, so it cannot fly at 10 mph. For this design, slow is the dangerous end: below about 45 mph a moderate 3 m/s gust can push the wing past stall.

- **Total part volume:** 215 cm³ (CAD). Printed mass is about 208 g at the recommended settings.
- **Print cost (PLA):** $2.10–6.00 for the whole airframe, depending on filament brand.
- **Angle of attack sitting level:** 0°. The wings have no built-in incidence and a symmetric airfoil, so they make no lift when level.
- **At 10 mph:** CL = 0 at 0° and about 0.53 at 10°. CD0 is about 0.031. Lift is under 6% of the aircraft's weight, so it stays planted on the ground.
- **Recommended flight speed without stabilisation:** 45–60 mph in light gusts. Below about 40 mph, gusts push it toward stall. Above about 60 mph, gust loads exceed 1.3 g.
- **Balance point (CG):** about 172–178 mm behind the nose tip. This estimate needs a glide test to confirm.

Geometry and assumptions

All figures below use the default parameters of the Onshape "F-35 RC Airframe" feature. Distances along the body are measured from the nose tip.

Quantity	Value
Length / wingspan	446 mm / 305 mm (1:35 of a real F-35A)
Wing reference area S (trapezoid carried to the centreline)	366 cm² (0.0366 m²)
Exposed wing area (outside the fuselage)	204 cm²
Root / tip chord, taper ratio	190 mm / 49 mm, λ = 0.26
Aspect ratio	2.54 (real F-35: 2.68)

Quantity	Value
Mean aerodynamic chord (MAC)	134 mm, leading edge at 189 mm
Airfoil, incidence	NACA 0006 symmetric, 0° to the fuselage axis
Horizontal tail / vertical tails	66 cm² exposed / 69 cm² (both fins)
Fuselage wetted area, fineness ratio	747 cm², L/d = 6.9
Open nozzle base	7.7 cm²

The flying-weight estimate assumes a typical small 50 mm EDF setup, since propulsion isn't chosen yet:

Item	Mass (g)
Printed airframe (PLA, settings in section 5)	208
3S 1000 mAh LiPo	85
50 mm EDF unit	55
30 A ESC	22
2 × 9 g servos	18
Wiring and connectors	12
Carbon spars, heat-set inserts, screws	12
Receiver	8
Total	**420**

At 420 g the wing loading is 115 g/dm² (37.7 oz/ft²). Small foam EDF jets usually sit around 50–80 g/dm², so this design is on the heavy side, and that drives the speed results below.

Volume by part

The five parts total 215 cm³ of solid CAD volume, which prints to about 208 g of PLA. The wings are the biggest single item by volume, but they print hollow with light infill, so they end up the lightest parts per cm³.

Part	Bounding box (mm)	CAD volume (cm^3)	Printed volume (cm^3)	Printed mass (g)
01 Crumple nose	91 × 39 × 36	10.5	10.5	13
02 Forward fuselage	202 × 96 × 45	58.2	58.2	72
03 Aft fuselage + tails	176 × 189 × 78	60.5	60.5	75
04 Right wing	146 × 105 × 9	42.9	19.3	24
05 Left wing	146 × 105 × 9	42.9	19.3	24
Total		**215.0**	**167.8**	**208**

The CAD volumes were measured in Onshape. The fuselage parts are already 0.8–1.2 mm shells, so they print essentially solid. Each wing panel's printed volume assumes two 0.4 mm walls (0.8 mm skin) plus 10% infill. Mass uses a PLA density of 1.24 g/cm^3.

Print cost (PLA)

One full airframe costs about $2.10–6.00 in PLA, depending on the brand. That allows 240 g of filament: the 208 g of parts plus about 15% for brims, supports under the fins and the canopy, and purge waste.

Filament tier (Oct 2026 street price)	$/kg	Airframe (240 g)	Spare crumple nose (15 g)
Budget single spool (US average)	$8.54	$2.05	$0.13
Mid-range (Elegoo PLA+, SUNLU)	~$13	$3.12	$0.20
Premium (Bambu PLA Basic, PolyTerra)	~$25	$6.00	$0.38

Electricity adds roughly $0.30–0.50 for about 18–22 h of printing, at about 0.1 kWh/h and $0.18/kWh. Because the nose is designed to be sacrificial, print 3–4 spare noses per batch. Prices are from Spoolhound's live PLA price tracker, checked today.

Print settings for low wing drag

Print each wing standing on its root face, so the layer lines run chordwise, in the same direction as the airflow. This orientation does more for drag than any slicer setting. If the wing lies flat instead, every layer step becomes a spanwise ridge facing the airflow.

Setting	Recommendation	Why
Orientation	Wing on its root face (104 mm tall), 5 mm brim, root insert hole facing down	Layer lines run streamwise. The leading edge prints as one smooth curve with no stair-steps.
Layer height	0.12 mm (0.08 mm if the printer handles it)	Smaller ridges. At these low Reynolds numbers, ridges facing the flow trip early separation.
Walls / skin	2 walls of 0.4 mm, outer wall printed first, outer wall at 30–40 mm/s	Thin, accurate skin with a cleaner surface
Seam	Aligned, placed on the trailing edge	Hides the seam bump at the one spot where flow already leaves the surface
Infill	8–10% gyroid; turn off top and bottom solid layers inside the wing	Light and stiff in torsion. Skipping solid layers prevents pillowing on the skin.
Spar channels	Add 2 extra walls around the 3.2 mm spar holes	The rods carry the wing load, so the holes need a solid ring of plastic around them.
Cooling / temperature	100% fan, nozzle 5 °C below your usual PLA temperature	Crisp trailing edge with less drooping or curling
Post-processing	Wet-sand 320 → 600, thin filler primer, 800-grit finish	Removes the remaining layer texture
Fuselage parts	0.16 mm layers, 3 walls (1.2 mm), 0% infill, tree supports only under the fins and canopy	The CAD shell is already the structure.

At your flight speeds (Reynolds number 40k–200k), a mirror-smooth wing can lose efficiency to laminar separation bubbles. The proven fix is a thin trip strip (0.2 mm tape) along 20–25% of the chord on the top surface. Once the wing is sanded smooth, test with and without it.

Angle of attack when sitting level

It is 0°. The wing chord lines are parallel to the fuselage axis (0° incidence), and NACA 0006 is symmetric, so its zero-lift angle is also 0°. A level model therefore makes zero lift until it rotates. This holds when the aircraft rests on its belly, and with landing gear only if the legs are sized to hold the fuselage axis level.

Two cheap ways to get a little lift without rotating:

- Make the nose gear 4–5 mm longer than the mains. That gives a 2–3° ground attitude, which makes the takeoff run noticeably shorter.
- Add a wing-incidence parameter of +1–2° to the feature. A small amount of built-in incidence keeps the fuselage closer to level in cruise. Real fighters use about 0–1°.

Lift and drag at 10 mph

At 10 mph the aircraft cannot fly. Lifting 420 g at that speed would need CL = 9.2, while a low-aspect-ratio wing like this one tops out near CL 0.9–1.0. The estimated stall speed is 30–34 mph, and at 10 mph the wing carries less than 6% of the weight even at 10° nose-up.

Flight conditions at 10 mph (4.47 m/s): dynamic pressure q = 12.2 Pa. The Reynolds number is 40,000 on the wing MAC and 133,000 on the fuselage length, so the boundary layer is fully laminar.

Angle of attack	CL	CD	Lift (gram-force)	Drag (gram-force)
0°	0.00	0.031	0	1.4
2°	0.11	0.033	5	1.5
4°	0.21	0.038	10	1.8
6°	0.32	0.048	15	2.2
8°	0.43	0.061	19	2.8
10°	0.53	0.078	24	3.6

How the estimate was made:

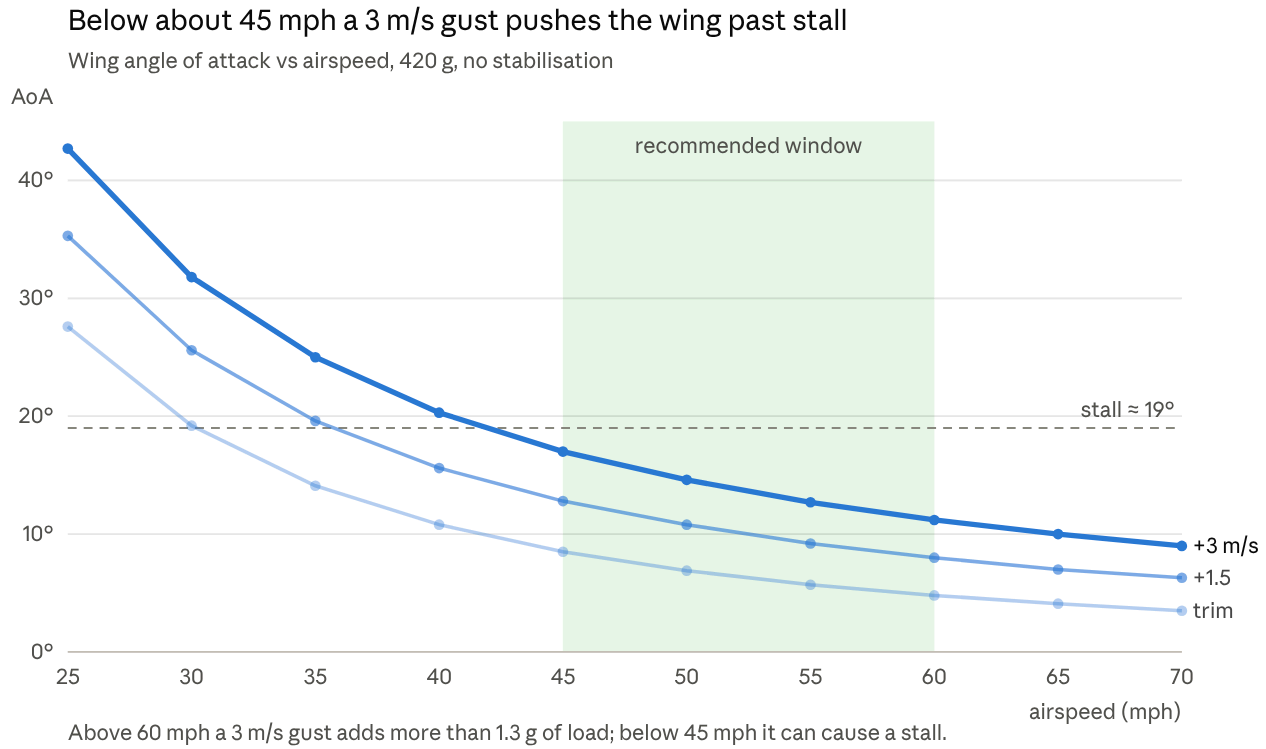
- **Lift slope:** Helmbold's low-aspect-ratio formula, CLα = 3.05/rad (0.053/°), stays linear up to about 18°.
- **Drag:** zero-lift drag CD0 = 0.031 is built from four pieces: laminar skin friction on wings, fuselage and tails (0.0085 + 0.0090 + 0.0080), +10% for interference and the screw holes, and base drag of the open nozzle (0.0027).
- **Induced drag:** $CL^2/(\pi \cdot 0.75 \cdot AR)$.
- **Uncertainty:** a real printed surface and laminar separation typically push CD0 up to 0.04–0.05 at these Reynolds numbers, so treat these values as a lower bound.

The formulas behind these numbers:

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C_L = \frac{2\pi A}{2+\sqrt{A^2+4}}\,\alpha \qquad C_D = C_{D0} +
\frac{C_L^2}{\pi e A} \qquad V_{stall} = \sqrt{\frac{2W}{\rho S C_{L,max}}}
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Gusts and turbulence: safe speed window

Without stabilisation, fly between 45 and 60 mph. Going slower is riskier for this design, not safer. At low speed the wing already needs a high angle of attack, and a vertical gust adds atan(w/V) on top. At 35 mph a 3 m/s gust lifts the wing from 14° to 25°, well past the roughly 19° stall.



Calculated: 420 g, S = 0.0366 m², CLα = 3.05/rad, sea-level air; gust Δα = atan(w/V)

The upper limit comes from loads rather than stall. A 3 m/s gust adds 0.78 g at 35 mph, 1.0 g at 45 mph and 1.33 g at 60 mph. Beyond 60 mph the light 1.2 mm shell and 3 mm spars start to work hard.

Side gusts on the ground are not a concern below about 30 mph:

- **Overturning:** with an 80 mm main-gear track, it takes a crosswind of roughly 36 mph to tip the aircraft over.
- **Lift on the ground:** at a 3° ground attitude, lift stays under 19% of weight up to 30 mph, so the wheels stay loaded.

If you need to run below the 45–60 mph window, these are the fixes, in order of effect:

1. Lower the weight.
2. Add a 3-axis gyro stabiliser (about 10 g).
3. Add wing area through chord rather than span, by lowering the trailing-edge forward sweep. The span is already at your 12 in limit.

Retractable landing gear (servo-driven)

I recommend tricycle gear like the real F-35. One 9 g servo drives all three legs through pushrods, and each leg locks over-centre so landing loads never reach the servo. The whole system adds about 45–60 g, which raises the stall speed by about 2 mph.

Leg	Position (from nose tip)	Retracts	Leg / wheel	Notes
Nose	x ≈ 105–120 mm, centreline	Forward, into the forward fuselage behind the firewall	2 mm music wire with a coil spring, 25 mm wheel	Must sit behind the firewall (94 mm) so the crumple nose can crush without hitting it. Use a nylon M2 screw as a breakaway fuse.
Mains (2)	x ≈ 190 mm (about 15° behind the CG), y = ±40 mm, giving an 80 mm track	Inward, into the belly where the wing stub meets the fuselage, like the real F-35	2 mm wire, 30 mm foam wheels	Mount plates attach with M2 × 4 heat-set inserts in printed bosses. Keep the wells clear of the spar rods at x = 238 and 264 mm.

Points to settle before modelling:

- **Space:** the inside of the fuselage is only about 33 mm tall at the wing root. Folded legs must fit in 30 mm with the wheel turned flat, so 25–30 mm wheels are the limit.
- **Ground angles:** make the nose leg 4–5 mm longer to sit at +2–3°. Allow 12° of rotation before the nozzle or stabilators strike the ground.
- **Servo linkage:** run the pushrods through printed guide tubes and set the travel so each leg reaches over-centre with about 1 mm of spare servo travel. A sequencer only matters if you add gear doors.
- **Alternative:** electric retract units (servoless, about 18–25 g each) are simpler to fit and fail-safe, but cost more and weigh about 15 g more in total.
- **Model support:** I can add gear-well pockets, mount bosses and pushrod channels to the FeatureScript as parameters once you pick the wheel size and retract hardware.

Recommended changes

The biggest single improvement is cutting weight. The other high-value fixes are control surfaces, the CG, and finishing the propulsion design. These are listed in order of impact:

1. **Print the wings and aft fuselage in LW-PLA.** It saves about 90 g. That brings the wing loading to about 88 g/dm² and the stall speed down to about 26–29 mph.
2. **Add control surfaces.** Elevons on the wing trailing edge plus all-moving stabilators, each with a parameter for hinge line and throw. Right now the aircraft has no way to control pitch or roll.
3. **Balance at 172–178 mm from the nose tip**, which is about 10–15% stability margin.
   - The neutral-point estimate is 191 mm. It includes a large destabilising fuselage term, so confirm the balance with hand-launched glide tests, starting at 172 mm.
   - The battery bay belongs in the forward fuselage, just behind the firewall.
4. **Pick the propulsion system.** A 50 mm EDF needs about 300–400 g of thrust for a 420 g jet. The current open nozzle (31 mm) would need intake ducts and a thrust tube. A pusher prop is the simpler alternative to test first.
5. **Re-run this analysis at the real weight.** I can rerun the speed window for your actual weight once the electronics are chosen.

Open question: what weight is the real electronics set? Every speed in this doc scales with the square root of the weight.